

Agentic GraphRAG for Financial Question Answering

Corporate financial reports, particularly SEC 10-K filings, contain extensive textual and numerical information spread across multiple sections, making accurate financial question answering a challenging retrieval and reasoning task. Conventional Retrieval-Augmented Generation (RAG) systems primarily depend on vector-based similarity search over unstructured passages, which often proves insufficient for complex queries requiring reasoning across multiple entities, fiscal years, and interconnected financial concepts. Graph-based RAG overcomes this by organizing information as structured knowledge, while agentic RAG further enhances retrieval by enabling a language model to plan, select, and coordinate retrieval strategies according to query needs. This project proposes an agentic GraphRAG system for financial question answering built on the FinReflectKG knowledge graph, that dynamically routes each query through lightweight or comprehensive retrieval based on a graph-connectivity and evidence-confidence complexity signal, rather than applying a fixed retrieval strategy uniformly. The system integrates graph traversal, hybrid document retrieval, and symbolic financial computation to retrieve evidence, perform calculations such as growth rates and ratios, and generate traceable responses. Preliminary evaluation on the FinReflectKG-MultiHop benchmark reveals a gap between retrieval quality and reasoning accuracy even when correct evidence is retrieved, which this work addresses through structured evidence formatting and adaptive routing. Answer provenance is scored at the individual claim level, verifying each generated claim against retrieved knowledge-graph triples rather than the answer as a whole. The framework targets improved temporal and cross-company reasoning while reducing computational overhead relative to uniformly comprehensive retrieval, with evaluation metrics spanning answer accuracy by reasoning type, retrieval efficiency, latency, and claim-level evidence traceability.